Virtualization

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Virtualization: An Introduction

- **Topic:** Virtualization
- **Description:** Explores the concept of virtualization and its importance in cloud computing.
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What is Virtualization?

- Separating hardware from software for improved system efficiency.
- Allows multiple operating systems (guest OS) to run on the same hardware, independent of the host OS.
- Enhances the use of compute engines, networks, and storage.
- Gartner Report (2009): Virtualization was a top strategic technology.

The Virtualization Layer

- Achieved by adding a *virtualization layer* between the hardware and the operating system.
- This layer is known as the **hypervisor** or **Virtual Machine Monitor (VMM)**.
- The VMM converts real hardware into virtual hardware.
- Enables different OS (e.g., Linux and Windows) to run simultaneously on the same physical machine.

Levels of Virtualization Implementation

- **Hardware Abstraction Level:** Virtualization directly on bare hardware.
  - Creates a virtual hardware environment for VMs.
  - Manages underlying hardware through virtualization.
  - Aims to improve hardware utilization by multiple users concurrently.
  - Example: Xen hypervisor for x86-based machines.

Levels of Virtualization Implementation (cont.)

- **Operating System Level:** Abstraction layer between the OS and user applications.
  - Creates isolated containers on a single physical server.
  - OS instances utilize the hardware.

Hypervisor Architecture

- Supports hardware-level virtualization on bare metal devices (CPU, memory, disk, network).
- Hypervisor sits directly between the physical hardware and the OS.
- Also known as VMM.

Types of Virtualization Architectures

- **Hypervisor Architecture:** Direct access to hardware.
- **Paravirtualization:** Guest OS is modified to cooperate with the hypervisor.
- **Host-Based Virtualization:** Virtualization layer runs on top of a host OS.

Examples of Virtualization Technologies

- **VMware Workstation:** Host-based virtualization.
- **Xen:** Hypervisor for IA-32, x86-64, Itanium, and PowerPC 970 hosts.
- **KVM (Kernel-based Virtual Machine):** Linux kernel virtualization infrastructure.
  - Supports hardware-assisted virtualization (Intel VT-x or AMD-v).
  - Supports paravirtualization (VirtIO framework).

Benefits of Virtualization

- Improved resource utilization.
- Application flexibility.
- Better system efficiency.
- Enlarged memory space (virtual memory).

Importance in Cloud Computing

- Virtualization is fundamental to cloud computing.
- Enables efficient resource allocation and management in cloud environments.
- Supports the creation of virtual machines and containers, which are essential building blocks for cloud services.